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Autonomous Robotic Tissue Palpation and Abnormalities Characterisation via Ergodic Exploration

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Abstract—We propose a novel autonomous robotic palpation framework for real-time elastic mapping during tissue exploration using a viscoelastic tissue model. The method combines force-based parameter estimation using a commercial force/torque sensor with an ergodic control strategy driven by a tailored Expected Information Density, which explicitly biases exploration toward diagnostically relevant regions by jointly considering model uncertainty, stiffness magnitude, and spatial gradients. An Extended Kalman Filter is employed to estimate viscoelastic model parameters online, while Gaussian Process Regression provides spatial modelling of the estimated elasticity, and a Heat Equation Driven Area Coverage controller enables adaptive, continuous trajectory planning. Simulations on synthetic stiffness maps demonstrate that the proposed approach achieves better reconstruction accuracy, enhanced segmentation capability, and improved robustness in detecting stiff inclusions compared to Bayesian Optimisation-based techniques. Experimental validation on a silicone phantom with embedded inclusions emulating pathological tissue regions further corroborates the potential of the method for autonomous tissue characterisation in diagnostic and screening applications.

Index Terms—Medical Robots and Systems; Sensor-based Control; Force and Tactile Sensing

[Figure 1 — placeholder]

Photograph of the experimental setup, arranged as a composite of three panels. **Bottom-left:** top-down view of a silicone phantom sample — a roughly square slab with a visible darker “Abnormality” inclusion near its centre. **Top-left:** the corresponding ground-truth elasticity map obtained via a dense grid scan, shown as a colour-coded 2D field with a vertical colour bar labelled *Elastic Modulus (kPa)*. **Right:** a Universal Robots UR3e manipulator shown palpating the silicone sample, with labelled call-outs identifying the *F/T Sensor* mounted at the wrist and the cylindrical *Indenter* tip in contact with the sample surface.

Fig. 1. Experimental setup for the proposed ergodic search: (bottom-left) silicone sample; (top-left) ground-truth elasticity map obtained using a grid-based approach; (right) the robotic manipulator palpating the silicone sample.

I. INTRODUCTION

PALPATION is a fundamental clinical procedure that allows physicians, by touch, to assess tissue stiffness, texture, and the presence of abnormalities such as tumours or swollen lymph nodes [1]. Its speed and non-invasiveness make it crucial for early diagnosis, particularly of conditions such as breast cancer. However, palpation is inherently subjective and sensitive to inter-practitioner variability, which may compromise diagnostic accuracy and repeatability for small or deep-seated lesions. For this reason, physical examinations are often complemented or replaced by imaging techniques such as mammography, the gold standard for early detection [2], and ultrasound [3].

Accurate assessment of tissue mechanics is equally critical in robot-assisted minimally invasive surgery (RMIS), where robots must adapt to variable stiffness for safe navigation and manipulation, especially for autonomous systems operating with minimal human supervision [4]. To support such needs, several technologies have emerged. Imaging-based elastography methods, e.g., ultrasound [5], magnetic resonance [6], or laser-based methods [7], estimate tissue stiffness but typically provide relative measures, are costly or slow, and cannot be easily integrated into RMIS. Tactile sensing, in contrast, offers real-time, quantitative stiffness measurements by capturing force and deformation. Previous works have shown its potential to reduce applied forces, shorten procedures, and improve surgical outcomes [8], [9]. Early stiffness mapping used uniform probing grids [10], [11], while later methods adopted Bayesian Optimisation (BO) to reduce the number of probing actions [12]–[14], often coupled with custom tactile sensors [8], [9], [15], guided by anatomical priors [16] or with ad-hoc refined segmentation strategies [15], [17].

In this work, we propose a method for autonomous robotic palpation that enables continuous estimation of elastic tissue stiffness using a viscoelastic contact model, relying solely on a commercial force/torque (F/T) sensor. The use of a viscoelastic model allows more accurate representation of tissue dynamics during palpation compared to purely elastic formulations, while abnormality characterisation in this work focuses on the estimated elastic stiffness. This sensing strategy provides a practical alternative to methods based on embedded or specialised sensors, which typically require customised hardware integration and lack compatibility with standard robotic platforms. In contrast, the use of an off-the-

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